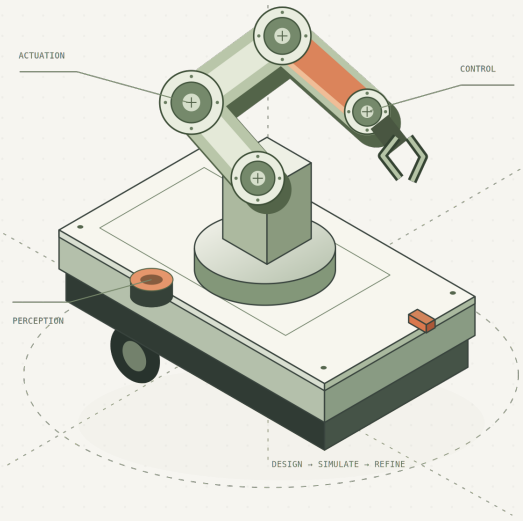
Nithyasri Rajulu

ENGINEERING PORTFOLIO

Mechanical design.
Motion control.
Machine perception.

M.Sc. Mechatronics student at KTH Royal Institute of Technology, with a background in Robotics and Automation and an interest in intelligent industrial machines.



CONCEPT ILLUSTRATION

Connecting mechanisms, control and perception.

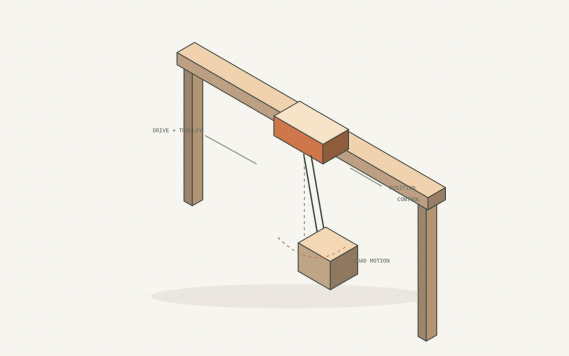
My projects span industrial drives, mechanical design, robot integration and computer vision. I’m seeking a master’s thesis with an industrial team in Sweden, starting in Spring 2027.

Thesis start: Spring 2027
Expected graduation: 14 June 2027
Location: Stockholm, Sweden

View my portfolio online | rajulu@kth.se

Mechanisms in motion.

Industrial projects in control, actuation and remote operation.



Concept illustration

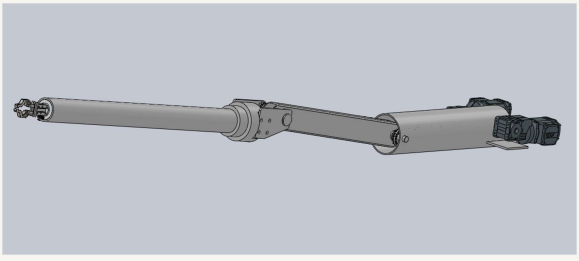
Crane anti-sway & servo control

Industrial control / SEW-Eurodrive

Crane anti-sway and radar servo-drive control, explored through modelling, controller design and drive configuration.

MATLAB · Python · MOVISUITE · MOVITOOLS

MATLAB and Python models for drive parameters and system response. · P and PI controller design for servo-drive behaviour. · Drive configuration and operator-interface development using MOVISUITE and MOVITOOLS.



CAD assembly from the supplied project report

Manipulator arm integration

Mechatronics / Visual Education Aids

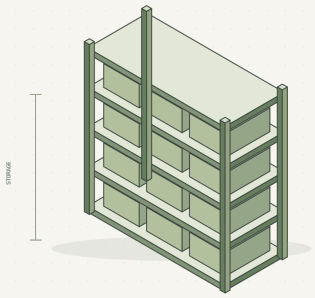
Mechanical and electrical integration for a remotely controlled manipulator arm.

SolidWorks · Stepper drives · Arduino · Zigbee

Mechanical layout and CAD integration of motors, gearing and mounts. · Stepper-motor and drive selection for the arm axes. · Arduino pulse-and-direction control with wireless joystick input.

From concept to mechanism.

Storage equipment, material handling and robotic end-effectors.



Concept illustration

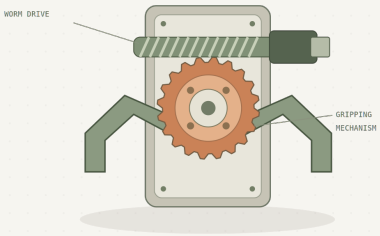
Material-handling product design

Product development / Prime Conveyors

Mechanical design for a vertical carousel storage unit and an AGV pallet.

CAD · Manufacturing drawings · DFMA

CAD modelling and manufacturing drawings for storage and transport equipment. · Component selection and design for manufacture and assembly. · Assembly-sequence planning and consideration of reusable components.



Concept illustration

Worm-drive gripper

Mechanical design / Team project

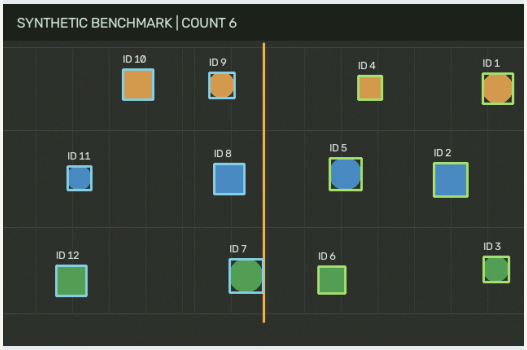
A worm-drive gripper concept for robotic part handling.

SolidWorks · Gear selection · Prototyping

Worm-and-worm-wheel arrangement and actuation concept. · SolidWorks models of the gripper components and assembly. · Mechanical interfaces, component selection and prototype planning.

Seeing the industrial task.

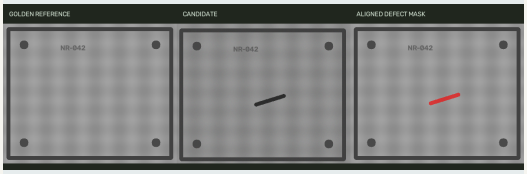
Image-processing demonstrations for counting, inspection and measurement.



Conveyor part counting

An OpenCV application for tracking parts and counting crossings along a conveyor.

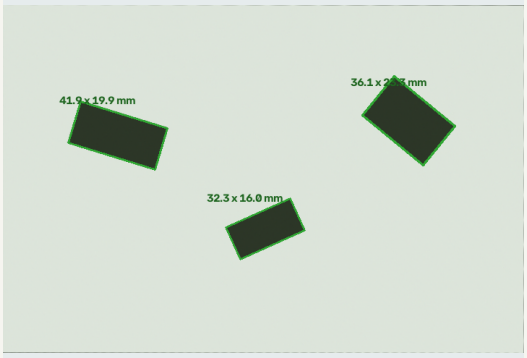
Python · OpenCV · Tracking



Surface defect inspection

An image-processing approach to finding surface differences against a reference image.

Python · OpenCV · Image registration



Perspective-calibrated metrology

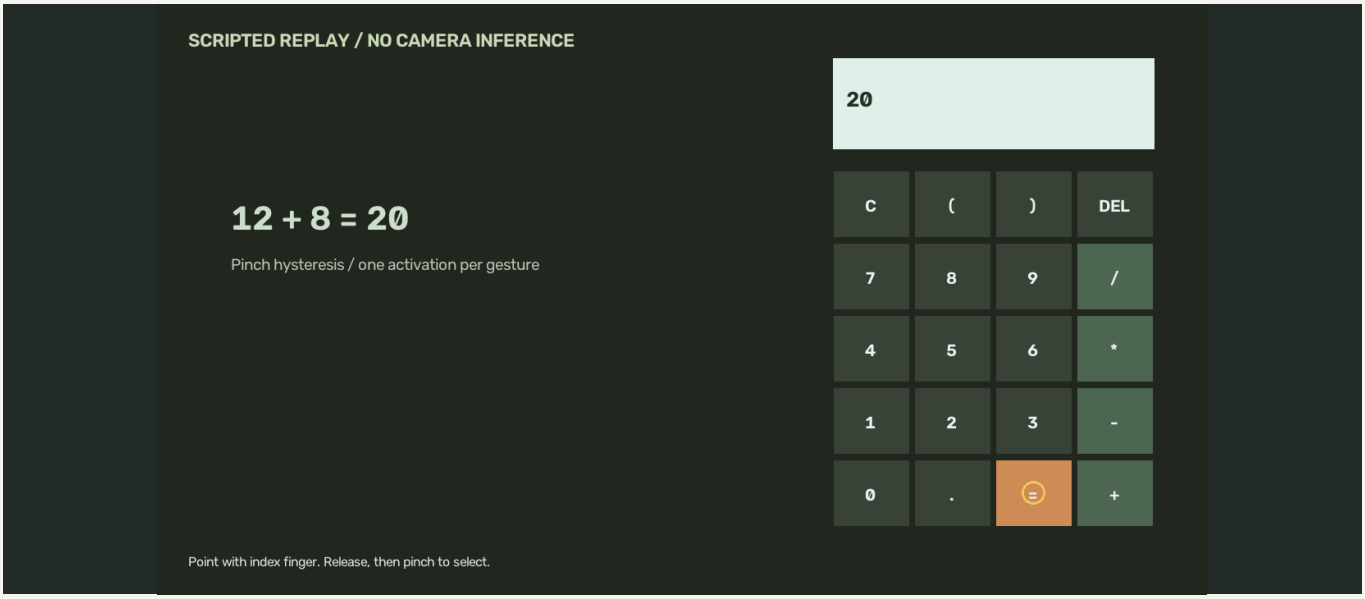
A computer-vision application for estimating planar part dimensions from an angled image.

Python · OpenCV · Homography

Images show generated demonstration scenes. Project descriptions focus on the image-processing approach.

From perception to interaction.

A calculator interface and ongoing robotics concepts.



SCRIPTED CALCULATOR INTERFACE DEMONSTRATION

Gesture-controlled virtual calculator

A gesture-controlled calculator interface using hand landmarks and pinch input.

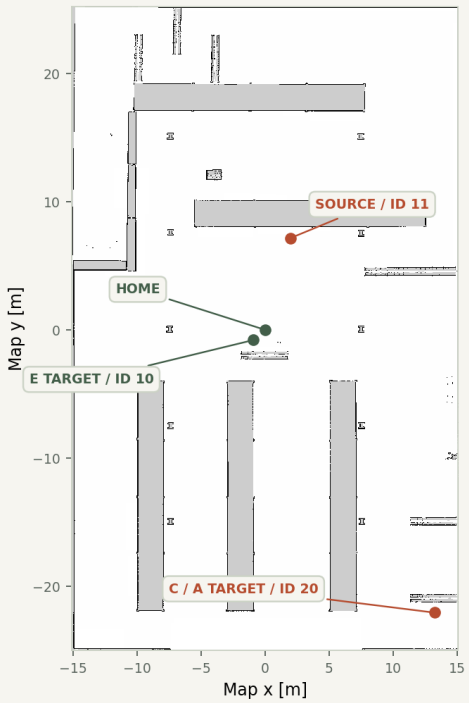
The application connects hand landmarks, fingertip targeting and pinch input with a virtual keypad. Its software includes arithmetic parsing, visual feedback and a mouse-input fallback.

Autonomous warehouse robot
An ongoing design and simulation project using Fusion 360, ROS 2, Nav2 and Gazebo to explore mobile material handling.

Warehouse scenario simulation
An ongoing Python project exploring vehicle movement and configurable obstacle scenarios, with AI and reinforcement learning as future directions.

Warehouse mobile manipulation.

ROS 2 Jazzy · Nav2 · py_trees · numerical inverse kinematics



MAP → TASK

Known standing poses

Home
x 0.00 / y 0.00 m

Source · ID 11
x 1.98 / y 7.15 m

E target · ID 10
x −0.90 / y −0.80 m

C / A target · ID 20
x 13.24 / y −22.07 m

Configured positions
No recorded trajectory

SUPPLIED OCCUPANCY MAP / CONFIGURED TASK POSITIONS

Coordinating navigation and manipulation

A ROS 2 course project combining mobile navigation, arm kinematics and vacuum pick-and-place.

- Nav2 action clients and sequential or behaviour-tree task logic.
- URDF-based forward kinematics, frame conversion and numerical inverse kinematics.
- Joint-trajectory commands, vacuum-gripper interfaces and AMCL localization concepts.

Course project within a supplied robotics framework. The focus is task logic, robot interfaces, coordinate transforms and motion planning in simulation.

Let’s define the next problem.

Thesis start: Spring 2027 | Expected graduation: 14 June 2027

I’m interested in a master’s thesis that connects practical engineering with robotics, perception or autonomous systems. My background brings together mechanical design, industrial drives and Python-based applications.

ROBOTICS & PRODUCTION

Mechanism design, actuator integration and automated assembly.

AUTONOMOUS SYSTEMS

Robot navigation, machine perception and scenario simulation.

MOBILE MACHINERY

Remote operation, motion control and automated material handling.

Nithyasri Rajulu

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B.E. Robotics and Automation, PSG College of Technology

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Portfolio website | LinkedIn | GitHub